

Management of Paracetamol Poisoning in Children's Emergency

General Facts

- Paracetamol is rapidly and completely absorbed in the small intestine, with peak serum concentration achieved within 4 hours of overdose of standard preparations.
- Metabolism of paracetamol takes place in the liver. Hepatotoxicity occurs due to a build-up of toxic metabolites after depleting liver stores of glutathione, which normally conjugates the metabolite and makes it harmless.
- Hepatotoxicity is unlikely if **<200mg/kg of paracetamol** in a child or **<10g** (20 of 500mg tablets) for an adult is taken unless they are at risk of hepatotoxicity.
 - The at-risk groups (hepatotoxicity from a lower paracetamol overdose) include those who are enzyme inducers (e.g. on certain antiepileptic drugs) and glutathione depleted states (e.g. those with eating disorders, cyanotic heart disease etc).
- Paracetamol overdose in the initial stage (<24h) are asymptomatic or have non-specific gastrointestinal symptoms. Hepatotoxicity and multi-organ failure tend to set in after 24h.
- The antidote N-acetylcysteine (NAC) is most effective when given **within 8 hours** of a single ingestion but may still be of benefit in those presenting later than this window period ¹.

Definitions

	Patients at risk for hepatotoxicity Δ	Patients not at risk for hepatotoxicity
Acute single ingestion	>150mg/kg or 7.5g (whichever lower)	>200mg/kg or 10g (whichever lower)
Massive single ingestion	>600mg/kg or >30g	

	<6yo <u>OR</u> with risk factors for hepatotoxicity Δ	>6yo <u>AND</u> with no risk factors for hepatotoxicity
Repeated supra-therapeutic ingestion	>100mg/kg/day or >4g/day (whichever is lower)	>150mg/kg/day or >6g/day (whichever is lower)

#NOTE: For obese children, this should be calculated based on ideal body weight

History

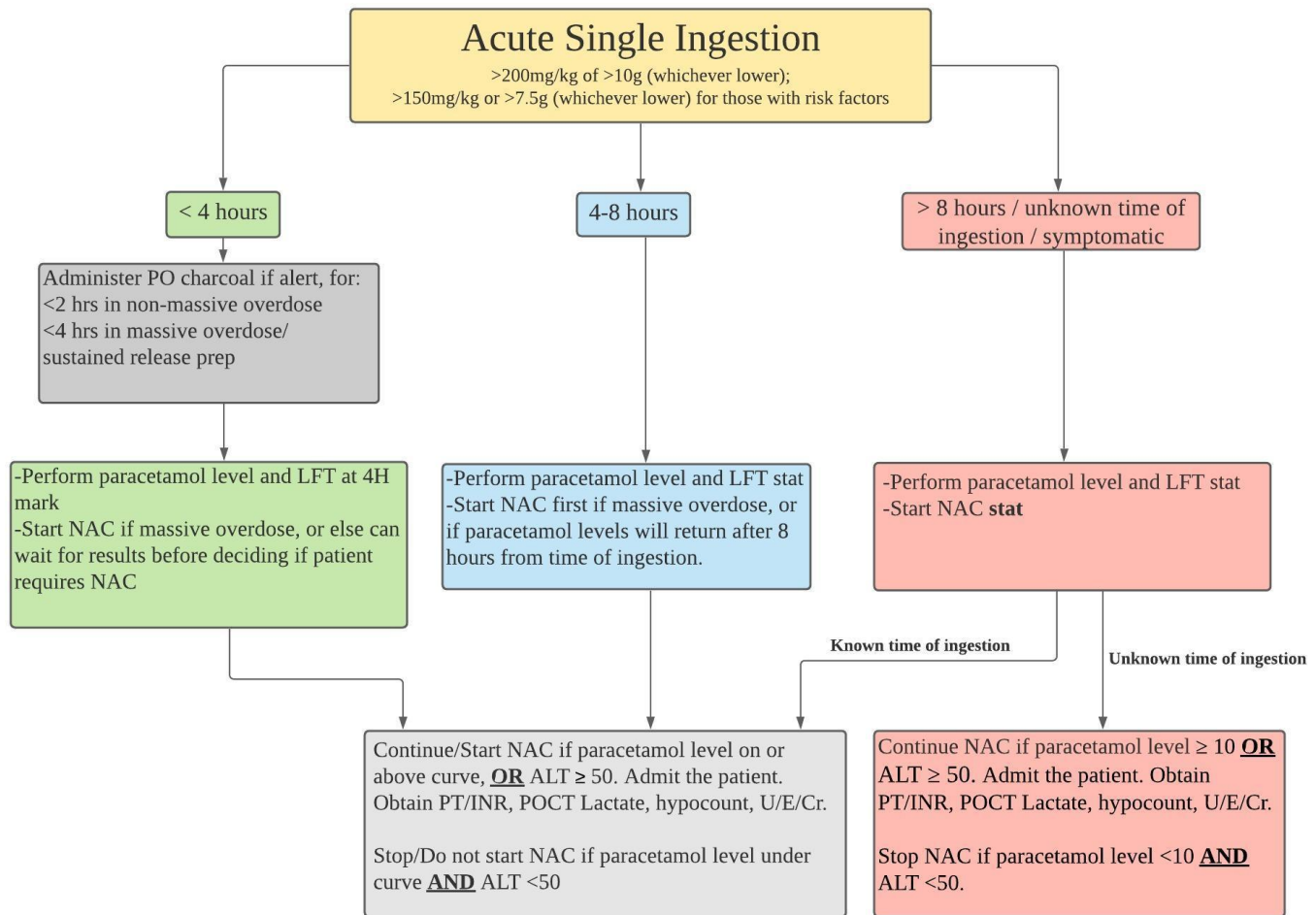
- Paracetamol ingestion history
 - Dose
 - Formulation (immediate vs modified release, tablet vs suspension), concentration
 - Time of ingestion
 - Single VS staggered overdose VS repeated supratherapeutic ingestion
 - Any co-ingestions
 - Symptoms eg: nausea, vomiting, abdominal pain
- Risk factors for hepatotoxicity Δ**
 - Pre-existing liver disease
 - Repeated excessive dosing
 - Chronic alcohol abuse
 - Enzyme-inducing drugs (e.g. barbiturates, carbamazepine, rifampicin, isoniazid)
 - Glutathione depletion states (e.g. malnutrition, prolonged fasting, anorexia nervosa)
- Suicide risk assessment

Principles in Management

- Ensure ABC stable. Resuscitation is rarely required unless there have been co-ingestions or massive paracetamol overdose with delayed presentation resulting in fulminant liver failure / severe lactic acidosis.
- Decontamination with activated charcoal 1g/kg (max 50g) can lower risk of hepatotoxicity².
 - Only applicable in single toxic ingestion AND in patients who are not obtunded.
 - For standard overdose, give activated charcoal within 2 hours of ingestion.
 - For massive overdose or sustained release formulations, give activated charcoal within 4 hours of ingestion.
- Antidote: Intravenous NAC. See below for dose and application in each category of overdose.

Disclaimer: This document reflects what is currently regarded as safe practice and does not replace the need for the application of clinical judgement to each individual presentation.

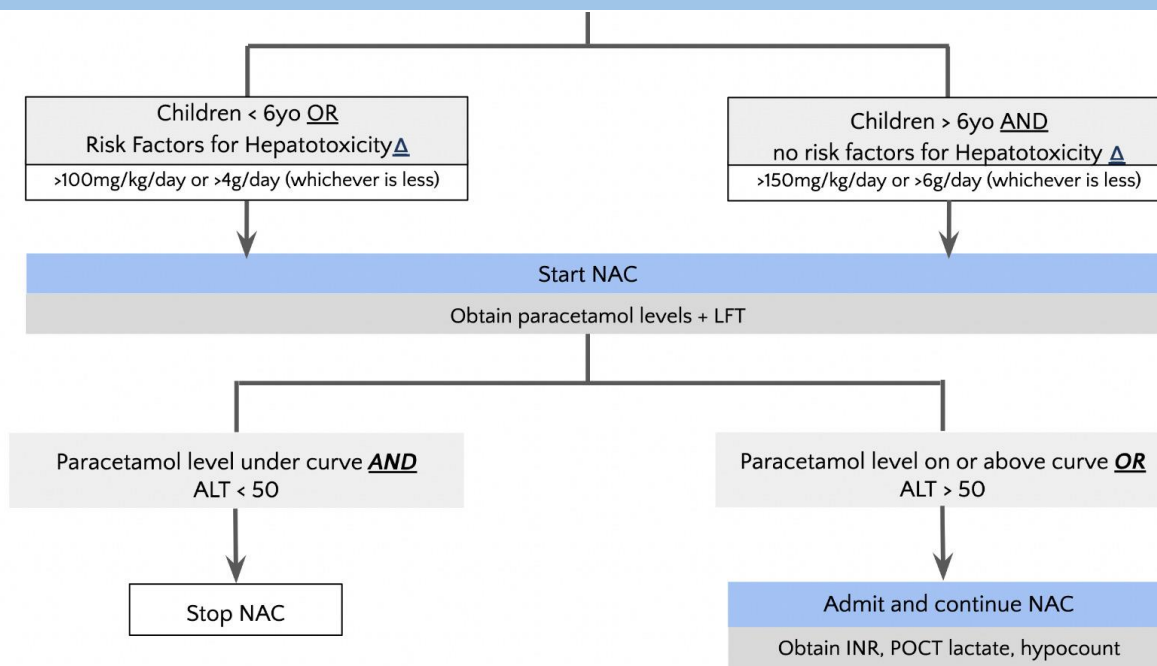
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- Activated charcoal should only be given in patients who are alert. Dose is 1g/kg, max of 50g.
- If the amount of paracetamol overdose taken does not exceed the thresholds stated above, check the paracetamol level and liver function test. There is no urgency in starting NAC unless the patient is symptomatic, which may suggest under-reporting of dose taken.
- It is recommended to perform paracetamol levels only from the 4-hour mark from time of single ingestion; this is due to falsely low levels if levels are done before that.
 - If levels are performed before the 4-hour mark and not reaching >150mg/L, a repeat level is required at the 4-hour mark in helping to decide if NAC is required.
 - If levels performed **before** the 4-hour mark is >150mg/L, continue/start NAC.
- Treatment with NAC started **within 8 hours** of a single ingestion will reduce severity of liver toxicity.
- If initial paracetamol level is ≥2x nomogram curve, consider doubling dose of 16h NAC infusion.

(B) Repeated Supratherapeutic Ingestion (RSI)

More than 1 dose of paracetamol ingested in the last 24-72 hours



- If multiple doses were taken but **not** reaching threshold of RSI, treat based on time of first ingestion using the acute single ingestion pathway above (A). If the first serum paracetamol level was measured within 2h of last ingested dose, repeat the level after 4hrs post ingestion to ensure there is no ongoing absorption.
- Δ Refer "History" for risk factors for hepatotoxicity

Administration of NAC

Dose of IV NAC and diluent required

Dose of IV NAC	Volume of diluent (D5%) required based on child's weight		
	<20 kg	20-50kg	>50kg
150mg/kg over 1 hour	3ml/kg	100ml	200ml
50mg/kg over 4 hours	7ml/kg	250ml	500ml
100mg/kg over 16 hours	14ml/kg	500ml	1000ml

- If initial paracetamol concentration was more than double the nomogram line, consider doubling the dose of the 16-hour infusion to 200mg/kg.
- It is recommended to perform laboratory investigations (to look for evidence of hepatic injury) after completion of the 21-hour infusion to decide if NAC should be continued.

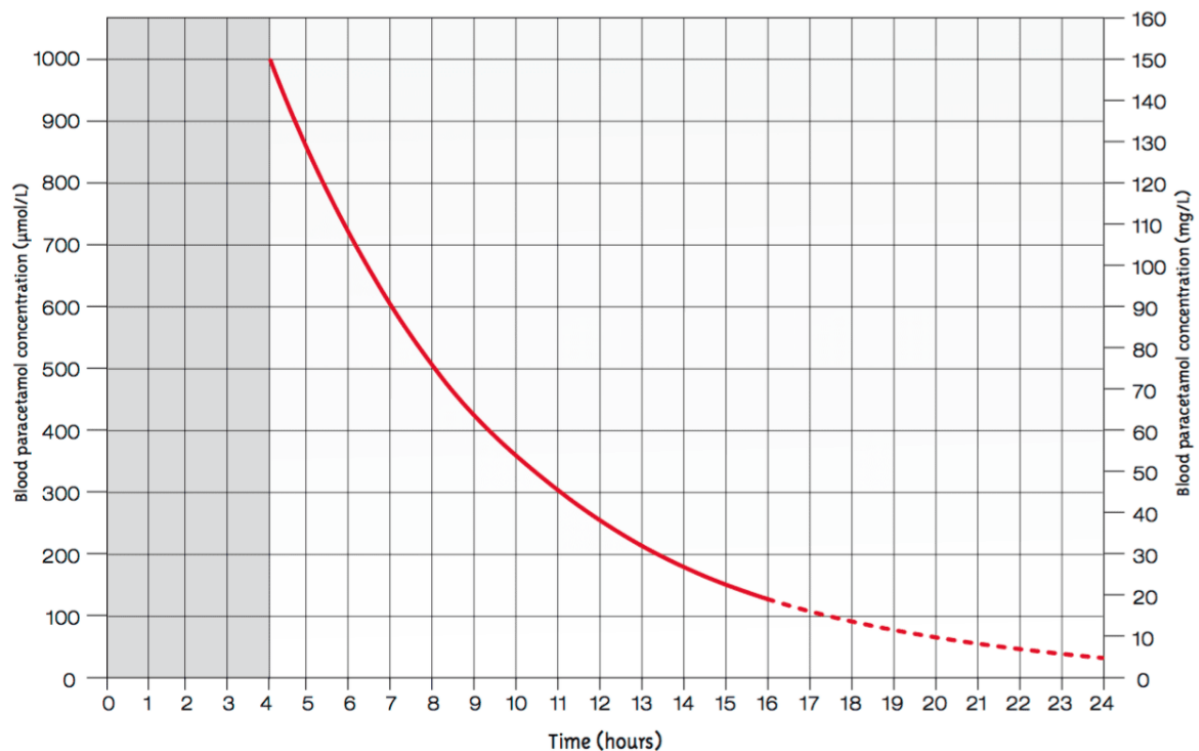
Anaphylactoid reactions to NAC

- Non-IgE mediated allergic reaction
- Occurs in about 10% of patients
- Management depends on severity of allergic manifestations:
 - Flushing: Monitor for development of allergic symptoms. Continue NAC if clinically indicated.
 - Mucocutaneous symptoms: Stop NAC infusion, administer IV diphenhydramine 0.5-1mg/kg (max dose 50mg), continue NAC if no further development of allergic reaction.
 - Airway or respiratory compromise (stridor / wheeze): Stop NAC infusion, administer nebulised adrenaline / salbutamol, IV diphenhydramine, IV hydrocortisone, IM adrenaline. Monitor for 1 hour before restarting IV NAC at a slower rate (half the infusion rate).
 - Hypotension: Stop and do not restart NAC. Manage as per anaphylaxis (IM adrenaline, IV hydrocortisone, IV diphenhydramine); give IV fluids for hypotension. Check with toxins expert regarding alternatives to NAC if paracetamol level or transaminitis is significant requiring treatment.

Severe Hepatotoxicity

- Consult Paediatrics GI/liver transplant team if features of fulminant liver failure from paracetamol overdose.

Appendix 1: Rumack-Matthew Nomogram



Note: Ensure correct units are used when utilising the nomogram — NUH lab's paracetamol level is in **mg/L** (i.e. use the value on the right of this nomogram)

References

1. Improved outcome of paracetamol-induced fulminant hepatic failure by late administration of acetylcysteine. Harrison PM, Keays R, Bray GP, Alexander GJ, Williams R. Lancet. 1990;335(8705):1572.
2. Massive paracetamol overdose: an observational study of the effect of activated charcoal and increased acetylcysteine dose (ATOM-2). Chiew AL, Isbister GK, Kirby KA, Page CB, Chan BSH, Buckley NA. Clin Toxicol (Phila). 2017;55(10):1055. Epub 2017 Jun 23.
3. Rumack BH, Matthew H. Acetaminophen poisoning and toxicity. Pediatrics 1975; 55:871-876
4. Uptodate: Clinical manifestations and diagnosis of acetaminophen poisoning in children and adolescents and management of acetaminophen poisoning in children and adolescents
5. MOH Clinical Practice Guidelines Dec/2011: Management of Poisoning
6. Daly, F. F., Fountain, J. S., Murray, L., Graudins, A., & Buckley, N. A. (2008). Guidelines for the management of paracetamol poisoning in Australia and New Zealand-explanation and elaboration. Medical Journal of Australia, 188(5), 296.